1. $f(x)$ defined at x_0
2. $\lim_{x \to x_0} f(x)$ exists
3. $f(x_0) = \lim_{x \to x_0} f(x)$

$$f(x) = \begin{cases} \frac{\sin 2x}{x} & x \neq 0 \\ 2 & x = 0 \end{cases}$$

Is $f(x)$ continuous?

$f(0) = 2$

$\lim_{x \to 0} \frac{\sin 2x}{x} = 2$

$f(0) = \lim_{x \to 0} \frac{\sin 2x}{x} = 2$

$$f(x) = \frac{x-2}{x-3}$$

$$f(x) = \begin{cases} x^2 & \text{if } x < 1 \\ 5 & \text{if } x = 1 \\ 3x+1 & \text{if } x > 1 \end{cases}$$

$\boxed{x = 1}$

$\lim_{x \to 1^+} (3x+1) = 4,$

$\lim_{x \to 1^-} x^2 = 1,$

$f(1) = 5,$

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$$f(x) = |x| = \begin{cases} x & x > 0 \\ -x & x < 0 \end{cases}$$

Differentiable function

$y = f(x)$

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

$$\lim_{h \to 0^+} \frac{(x+h) - x}{h} = 1$$

$$\lim_{h \to 0^-} \frac{-(x+h) + x}{h} = -1$$

$f(x) = x^2$

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

$f(x+h) = (x+h)^2 = x^2 + 2hx + h^2$

$$f'(x) = \lim_{h \to 0} \frac{\cancel{x^2} + 2hx + h^2 - \cancel{x^2}}{h} = \lim_{h \to 0} \frac{2x+h}{1} = 2x$$

$$f(x) = \begin{cases} x & x > 0 \\ -x & x < 0 \end{cases}$$

$$f(x+h) = \begin{cases} x+h & x > 0 \\ -(x+h) & x < 0 \end{cases}$$

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$f(x)$	$f'(x)$	y' $f'(x)$ $\frac{dy}{dx}$ $\frac{df}{dx}$
x^n	$n x^{n-1}$	
$ax^2 + bx + c$	$2ax + b$	
c	0	
$(g(x))^n$	$n (g(x))^{n-1} \cdot g'(x)$	
$\sqrt{g(x)}$	$= \frac{g'(x)}{2\sqrt{g(x)}}$	
$\sqrt{x}$	$\frac{1}{2\sqrt{x}}$	
$g(x)h(x)$	$g(x)h'(x) + g'(x)h(x)$	
$\frac{g(x)}{h(x)}$ $h(x) \neq 0$	$\frac{g'(x)h(x) - h'(x)g(x)}{(h(x))^2}$	

1) $f(x) = 2x^3 + 3x - 2$ $\quad f'(x) = 6x^2 + 3$

2) $f(x) = (2x^2 - 5x)^3$ $\quad f'(x) = 3(2x^2 - 5x)^2 (4x - 5)$

3) $y = \sqrt{2x-1}$ $\quad y' = \frac{2}{2\sqrt{2x-1}}$

4) $\frac{x-2}{x^2-3}$ $\quad y' = \frac{(x^2-3) - (x-2)2x}{(x^2-3)^2}$

5) $y = (2x-2)(x+3)$ $\quad y' = (2x-2) + (x+3)2$

$$\lim_{h \to 0} \frac{(x+h)+x}{h} = -1$$

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$f(x)$	$f'(x)$	y' $f(x)$ $\frac{dy}{dx}$ $\frac{df}{dx}$
x^n	$n x^{n-1}$	
ax^2+bx+c	$2ax+b$	
c	0	
$(g(x))^n$	$n(g(x))^{n-1} \cdot g'(x)$	
$\sqrt{g(x)}$	$= \frac{g'(x)}{2\sqrt{g(x)}}$	
$\sqrt{x}$	$\frac{1}{2\sqrt{x}}$	
$g(x)h(x)$	$g(x)h'(x)+g'(x)h(x)$	
$\frac{g(x)}{h(x)}$ $h(x) \neq 0$	$\frac{g'(x)h(x)-h'(x)g(x)}{(h(x))^2}$	

1) $f(x) = 2x^3+3x-2$ $\quad f'(x) = 6x^2+3$

2) $f(x) = (2x^2-5x)^3$ $\quad f'(x) = 3(2x^2-5x)^2(4x-5)$

3) $y = \sqrt{2x-1}$ $\quad y' = \frac{2}{2\sqrt{2x-1}}$

4) $y = \frac{x-2}{x^2-3}$ $\quad y' = \frac{(x^2-3)-(x-2)2x}{(x^2-3)^2}$

5) $y = (2x-2)(x+3)$ $\quad y' = (2x-2)+(x+3)2$

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$f(x)$	$f'(x)$	y' $f(x)$ $\frac{dy}{dx}$ $\frac{df}{dx}$
$\sin x$	$\cos x$	
$\cos x$	$-\sin x$	
$\tan x$	$\sec^2 x$	
$\sec x$	$\sec x \tan x$	
$\csc x$	$-\csc x \cot x$	
$\cot x$	$-\csc^2 x$	
$\sin g(x)$	$\cos g(x) \cdot g'(x)$	
$\cos g(x)$	$-\sin g(x) \cdot g'(x)$	

1) $f(x) = 2x^3+3x-2$ $\quad f'(x) = 6x^2+3$

2) $f(x) = (2x^2-5x)^3$ $\quad f'(x) = 3(2x^2-5x)^2(4x-5)$

3) $y = \sqrt{2x-1}$ $\quad y' = \frac{2}{2\sqrt{2x-1}}$

4) $y = \frac{x-2}{x^2-3}$ $\quad y' = \frac{(x^2-3)-(x-2)2x}{(x^2-3)^2}$

5) $y = (2x-2)(x+3)$ $\quad y' = (2x-2)+(x+3)2$

$y = \sin(2x+3)$ $\quad y' = \cos(2x+3)(2)$

$y = \csc(5x^2-2)$ $\quad y' = -\csc(5x^2-2)\cot(5x^2-2) \cdot 10x$

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$f(x)$	$f'(x)$	y' $\frac{dy}{dx}$	$f'(x)$ $\frac{df}{dx}$
a^x	$a^x \ln a$		
e^x	e^x		
$a^{f(x)}$	$a^{f(x)} f'(x) \ln a$		
$e^{f(x)}$	$e^{f(x)} f'(x)$		

$f(x)$	$f'(x)$
$\ln x$	$\frac{1}{x}$
$\ln f(x)$	$\frac{1}{f(x)} f'(x)$

$y = e^{5x^2-7}$ $\quad y' = e^{5x^2-7}(10x)$

$y = 2^{\sin x}$ $\quad y' = 2^{\sin x} \cos x \ln 2$

$y = \pi^{x^2}$ $\quad y' = \pi^{x^2} (2x) \ln \pi$

$y = \ln(5x+7)$ $\quad y' = \frac{5}{5x+7}$

$y = \ln 6x$ $\quad y' = \frac{1}{x}$

$= \ln 6 + \ln x$

$y = x^x$ $\quad y' = x^x[1+\ln x]$

$\ln y = x \ln x$

$\frac{1}{y} y' = x\frac{1}{x} + \ln x$

$= 1 + \ln x$

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$f(x)$	$f'(x)$
$\ln x$	$\frac{1}{x}$
$\ln f(x)$	$\frac{1}{f(x)} f'(x)$

$y = f(x)$

$f(x,y) = 0$

Find the first derivative for the following functions

(1) $2xy^2 + ye^x = xy$

$2[y^2 + x(2yy')] + [y'e^x + ye^x] = [y + y'x]$

$4xyy' + y'e^x - y'x = y - 2y^2 - ye^x$

$y'[4xy + e^x - x] =$

$y = \ln(5x+7)$ $\quad y' = \frac{5}{5x+7}$

$y = \ln 6x$ $\quad y' = \frac{1}{x}$

$= \ln 6 + \ln x$

$y = x^x$ $\quad y' = x^x[1+\ln x]$

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$= 1 + \ln x$

CS CamScanner

$f(x)$	$f'(x)$	y' $f(x)$ $\frac{dy}{dx}$ $\frac{df}{dx}$
a^x	$a^x \ln a$	
e^x	e^x	
$a^{f(x)}$	$a^{f(x)} f'(x) \ln a$	
$e^{f(x)}$	$e^{f(x)} f'(x)$	

$y = e^{5x^2-7}$ $\quad y' = e^{5x^2-7}(10x)$

$y = 2^{\sin x}$ $\quad y' = 2^{\sin x} \cos x \ln 2$

$y = \pi^{x^2}$ $\quad y' = \pi^{x^2} (2x) \ln \pi$

$f(x)$	$f'(x)$
$\ln x$	$\frac{1}{x}$
$\ln f(x)$	$\frac{1}{f} f'(x)$

$y = \ln(5x+3)$ $\quad y' = \frac{5}{5x+3}$

$y = \ln 6x$ $\quad y' = \frac{1}{x}$

$= \ln 6 + \ln x$

$y = x^x$ $\quad y' = x^x[1+\ln x]$

$\ln y = x \ln x$

$\frac{1}{y} y' = x \frac{1}{x} + \ln x$

$= 1 + \ln x$

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$f(x)$	$f'(x)$
$y = f(y)$	
$f(x,y) = 0$	

Find the first derivative for the following functions

(1) $2xy^2 + ye^x = xy$

$2[y^2 + x(2yy')] + [y'e^x + ye^x] = (y + y'x)$

$4xyy' + y'e^x - y'x = y - 2y^2 - ye^x$

$y'[4xy + e^x - x] =$

$f(x)$	$f'(x)$
$\ln x$	$\frac{1}{x}$
$\ln f(x)$	$\frac{1}{f(x)} f'(x)$

$y = \ln(5x+3)$ $\quad y' = \frac{5}{5x+3}$

$y = \ln 6x$ $\quad y' = \frac{1}{x}$

$= \ln 6 + \ln x$

$y = x^x$ $\quad y' = x^x[1+\ln x]$

$\ln y = x \ln x$

$\frac{1}{y} y' = x \frac{1}{x} + \ln x$

$= 1 + \ln x$

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$y = \sinh x = \frac{e^x - e^{-x}}{2}$

$y' = \frac{1}{2}[e^x + e^{-x}] = \frac{e^x + e^{-x}}{2} = \cosh x$

$y = \cosh x = \frac{e^x + e^{-x}}{2}$

$y' = \frac{e^x - e^{-x}}{2} = \sinh x$

$y = \operatorname{Csech} x = \frac{2}{e^x - e^{-x}}$

$y' = \frac{-2(e^x + e^{-x})}{(e^x - e^{-x})^2} = -\frac{(e^x + e^{-x})}{(e^x - e^{-x})} \cdot \frac{1}{\frac{(e^x - e^{-x})}{2}}$

$= -\operatorname{Cth} x \cdot \operatorname{Csech} x$

$f(x)$	$f'(x)$
$\ln x$	$\frac{1}{x}$
$\ln f(x)$	$\frac{1}{f(x)} f'(x)$

$y = \ln(5x+3)$ $\quad y' = \frac{5}{5x+3}$

$y = \ln 6x$ $\quad y' = \frac{1}{x}$

$6 + \ln x$

$y = x^x$ $\quad y' = x^x[1+\ln x]$

$\ln x \ln x$

$= x \frac{1}{x} + \ln x$

$= 1 + \ln x$

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$y = \cos^{-1} f(x)$

$f(x) = \cos y$

$f'(x) = -\sin y \; y'$

$y' = \dfrac{-f'(x)}{\sin y} = -\dfrac{f'(x)}{\sqrt{1-(f(x))^2}}$

$\sin^2 y + \cos^2 y = 1$

$\sin^2 y + (f(x))^2 = 1$

$\sin^2 y = 1 - (f(x))$

$\sin y = \sqrt{1-(f(x))^2}$

$f(x) = \tan^{-1} e^x$

$f'(x) = \dfrac{e^x}{1+e^{2x}}$

$f(x) = \sin^{-1} 2x$

$f'(x) = \dfrac{\frac{1}{x}}{\sqrt{1-(2x)^2}}$

$y = \sin^{-1} x$

$x = \sin y$

$1 = \cos y \; y'$

$y' = \dfrac{1}{\cos y} = \dfrac{1}{\sqrt{1-x^2}}$

$\sin^2 y + \cos^2 y = 1$

$x^2 + \cos^2 y = 1$

$\cos^2 y = 1 - x^2$

$\cos y = \sqrt{1-x^2}$

$y = \tan^{-1} x$

$x = \tan y$

$1 = \sec^2 y \; y'$

$y' = \dfrac{1}{\sec^2 y} = \dfrac{1}{1+x^2}$

$1 + \tan^2 y = \sec^2 y$

$1 + x^2 = \sec^2 y$

$$x = x(t) = \sin t$$
$$y = y(t) = \cos t$$

$$\frac{dy}{dx} = \frac{dy}{dt}\,\frac{dt}{dx} = -\frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{\dot{y}}{\dot{x}} = \frac{-\sin t}{\cos t} = -\tan t$$

$$\frac{d^2y}{dx^2} = -\frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{d}{dx}\left(\frac{\dot{y}}{\dot{x}}\right)$$

$$= \frac{d}{dt}\,\frac{dt}{dx}\left(\frac{\dot{y}}{\dot{x}}\right)$$

$$= \frac{d/dt\left(\frac{\dot{y}}{\dot{x}}\right)}{\dot{x}} = \frac{1}{\dot{x}}\left[\frac{\ddot{y}\,\dot{x} - \ddot{x}\,\dot{y}}{\dot{x}^2}\right]$$

$$= \frac{1}{\cos^2 t}\left[-\cos^2 t - (-\sin t)(-\sin t)\right]$$

$$= \frac{-1}{\cos^2 t}\left[\cos^2 t + \sin^2 t\right] = \frac{-1}{\cos^2 t}$$

$x = \sin t$ $\qquad$ $y = \cos t$

$\dot{x} = \cos t$ $\qquad$ $\dot{y} = -\sin t$

$\ddot{x} = -\sin t$ $\qquad$ $\ddot{y} = -\cos t$

$$\lim_{x\to 0}\frac{\sin x}{x} = 1$$

$$\left|\begin{array}{ll} f(x)=\sin x & f(0)=0 \\ f'(x)=\cos x & f'(0)=1 \\ f''(x)=-\sin x & f''(0)=0 \\ f^{iv}(x)=-\cos x & f'''(0)=-1 \end{array}\right.$$

$$f(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4 + a_5 x^5 + ___ \qquad f(0)=a_0$$

$$f'(x) = a_1 + 2a_2 x + 3a_3 x^2 + 4a_4 x^3 + 5a_5 x^4 + __ \qquad f'(0)=a_1$$

$$f''(x) = 2a_2 + 3.2\, a_3 x + 4.3\, a_4 x^2 + 5.4\, a_5 x^3 + __ \qquad f''(0)=2a_2$$

$$f'''(x) = 3.2\, a_3 + 4.3.2\, a_4 x + 5.4.3.\, a_5 x^2 + \cdots \qquad f'''(0)=3.2\, a_3$$

$$f^{(4)}(x) = 4.3.2\, a_4 + 5.4.3.2\, a_5 x + __ \qquad f^{(4)}(0)=4.3.2\, a_4$$

$$f(x) = \cancel{f(0)} + f'(0)x + \frac{\cancel{f''(0)}}{2}x^2 + \frac{f'''(0)}{3!}x^3 + \frac{\cancel{f^{(4)}(0)}}{4!}x^4 + __$$

$$\sin x = x - \frac{1}{3!}x^3 + \frac{1}{5!}x^5 - \frac{1}{7!}x^7 + ___ = x\left[1 - \frac{1}{3!}x^2 + \frac{1}{5!}x^4 - __\right]$$

$$\lim_{x\to 0}\frac{\sin x}{x}=1$$

$$\begin{array}{ll} f(x)=\sin x & f(0)=0 \\ f'(x)=\cos x & f'(0)=1 \\ f''(x)=-\sin x & f''(0)=0 \\ f^{(3)}(x)=-\cos x & f^{(3)}(0)=-1 \end{array}$$

$$\cos x = 1-\frac{1}{2!}x^2+\frac{1}{4!}x^4$$

$$\lim_{x\to 0}\frac{1-\cos x}{x^2}$$

$$\begin{array}{ll} f(x)=\cos x & f(0)=1 \\ f'(x)=-\sin x & f'(0)=0 \\ f''(x)=-\cos x & f''(0)=-1 \\ f'''(x)=\sin x & f'''(0)=0 \\ f^{(4)}(x)=\cos x & f^{(4)}(0)=1 \end{array}$$

$$1-\cos x = \left[x-\frac{1}{2!}x^2+\frac{1}{4}x^4-\cdots\right]$$

$$=\frac{1}{2!}x^2-\frac{1}{4!}x^4+\cdots$$

$$\cos x = x+\cdots$$

$$f(x)=\cancel{f(0)}+f'(0)x+\frac{\cancel{f''(0)}}{2}x^2+\frac{f'''(0)}{3!}x^3+\frac{\cancel{f^{(4)}(0)}}{4!}x^4+\cdots$$

$$\sin x = x-\frac{1}{3!}x^3+\frac{1}{5!}x^5-\frac{1}{7!}x^7+\cdots = x\left[1-\frac{1}{3!}x^2+\frac{1}{5!}x^4-\cdots\right]$$